



# Molecular dynamics simulation of strengthening dependence on precipitate Cr composition in Fe-15at.%Cr alloy

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## ABSTRACT

Molecular Dynamics simulation was employed to study precipitate composition dependence on strengthening. Edge dislocation interaction with pure, 80at.%, and 60at.% Cr precipitates of different sizes in a matrix of Fe-15at.%Cr was investigated. The precipitates were found to be relatively hard. This is evident from the absence of shearing mechanism after the dislocation has bypassed them, the formation of an Orowan-like dislocation shape, and comparatively higher stress values. Precipitates with higher Cr content were found to greatly impede dislocation glide as indicated by the time taken by the dislocation to bypass them. The composition dependence on critical unpinning stress was also observed. The interaction of edge dislocation with precipitates having higher Cr composition leads to higher critical unpinning stress. The extent of critical unpinning stress dependence on precipitate composition is, however, not as high as was reported previously. Besides, the study has also confirmed the fact that  $\alpha'$  precipitation results in the hardening of high Cr ferritic/martensitic steels.

## 1. Introduction

While nuclear power remains a viable alternative to the world's growing energy demand, the continuous operation of nuclear power plants greatly depends on durable structural materials that can withstand severe radiation environments. Fe-Cr based Ferritic/Martensitic (FM) steels are considered as potential materials that will be very suitable for use in both current and future nuclear reactors. This is due to their high thermal conductivity, lower coefficient of thermal expansion, better void swelling resistance and high resistance to corrosion (Idhil et al., 2012; Van den Bosch et al., 2013; Yvon and Carré, 2009; Zinkle and Busby, 2009; Klueh and Harries, 2001). FM steels containing high Chromium (Cr) content, with a composition of around 9–14 at.%Cr, are commonly used for industrial applications that require high strength at high temperatures similar to that obtained in nuclear power plants (Bonny et al., 2010).

Materials degradation associated with neutron irradiation in nuclear fission reactors is a major source of concern. Radiation causes significant damage to these alloys, such as radiation-induced segregation of Cr at defect sinks, clustering of Cr atoms into Cr-rich  $\alpha'$  precipitate, irradiation hardening, as well as other damages, thereby affecting their

mechanical properties (Chen et al., 2017).

Mechanical properties of Fe-Cr alloys rely heavily on the atomic distribution of its constituent atoms and can be degraded by phenomena such as Cr precipitation (Grammatikopoulos and Nordlund, 2014). Studies have shown that Fe-Cr alloys containing high Cr composition (over 9at.% Cr) are susceptible to the precipitation of the bcc Cr-rich  $\alpha'$  phase (Reese et al., 2018). The formation of  $\alpha'$  precipitates increases radiation-induced strengthening above 9 %Cr (Terentyev et al., 2013a).

The development of transmission electron microscopy enabled the observation of dislocation. This paved the way for the design of alloys with appropriate strength and ductility using precipitation hardening (Keyhani and Roumina, 2017). This is because, the motion of dislocation results in macroscopic plastic deformation in materials, and obstacles such as precipitates were found to impede the motion of dislocation thereby leading to the strengthening of these materials.

MD studies, such as (Shim et al., 2009), were conducted to investigate the effect of Cr solute atoms and bcc Cr-rich  $\alpha'$  precipitates on edge dislocation glide in bcc Fe. This is to help understand the effect of  $\alpha'$  formation on the mechanical properties of high Cr FM steels. Solid solutions of Fe-Cr containing non-interacting Cr atoms were found not

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